Yunho Choi

Contact Information

Ph.D. Student
Robot Learning Laboratory
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Research Interests

Vision-based Robot Control, Embodied AI, Reinforcement Learning

Education

Ph.D. in Electrical and Computer Engineering Mar. 2017 - Present

- Seoul National University, Seoul, Korea
- Advisor: Prof. Songhwai Oh GPA: 4.03 / 4.3

B.S. in Electrical and Computer Engineering Mar. 2013 - Feb. 2017

- Seoul National University, Seoul, Korea GPA: 3.91 / 4.3

Publications

Yunho Choi, Jaeseok Heo, Nuri Kim, and Songhwai Oh, **"KG-Nav: Bridging the Gap between Visual Servoing and Image-Goal Navigation via Keypoint Graph-Based Reinforcement Learning,"** *Under Review*

Hwiyeon Yoo, **Yunho Choi**, Jeongho Park and Songhwai Oh, **"Commonsense-Aware Object Value Graph Navigation for ObjectNav,"** *Under Review*

Nuri Kim, Obin Kwon, Hwiyeon Yoo, **Yunho Choi**, Jeongho Park, and Songhwai Oh, **"Topological Semantic Graph Memory for Image-Goal Navigation,"** in *Proc of the Conference on Robot Learning (CoRL)*, Dec. 2022. (Oral Presentation, Acceptance Rate: 6.5%)

Obin Kwon, Nuri Kim*, **Yunho Choi***, Hwiyeon Yoo*, Jeongho Park*, and Songhwai Oh, **"Visual Graph Memory with Unsupervised Representation for Visual Navigation,"** in *Proc. of the International Conference on Computer Vision (ICCV)*, Oct. 2021.

Yunho Choi and Songhwai Oh, **"Image-Goal Navigation via Keypoint-Based Reinforcement Learning,"** in *Proc. of the International Conference on Ubiquitous Robots (UR)*, Jul. 2021.

Yoonseon Oh, Kyunghoon Cho, **Yunho Choi**, and Songhwai Oh, "Chance-Constrained Multi-Layered Sampling-Based Path Planning for Temporal Logic-Based Missions," *IEEE Transactions on Automatic Control (TAC)*, Dec. 2020.

Kyungjae Lee, Jaegu Choy, **Yunho Choi**, Hogun Kee, and Songhwai Oh, "No-Regret Shannon Entropy Regularized Neural Contextual Bandit Online Learning for Robotic Grasping," in *Proc. of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Oct. 2020.

Kyungjae Lee, **Yunho Choi**, and Songhwai Oh, "Inverse Optimal Control from Demonstrations with Mixed Qualities," in *Proc. of the International Conference on Ubiquitous Robots (UR)*, Jun. 2020.

Yunho Choi, Hogun Kee, Kyungjae Lee, Jaegoo Choy, Junhong Min, Sohee Lee, and Songhwai Oh, **"Hierarchical 6-DoF Grasping with Approaching Direction Selection,"** in *Proc. of the IEEE International Conference on Robotics and Automation (ICRA)*, May 2020.

Yunho Choi, Kyungjae Lee, and Songhwai Oh, **"Distributional Deep Reinforcement Learning with a Mixture of Gaussians,"** in *Proc. of the IEEE International Conference on Robotics and Automation (ICRA)*, May. 2019.

Hyemin Ahn, Timothy Ha*, **Yunho Choi***, Hwiyeon Yoo*, and Songhwai Oh, "Text2Action: Generative Adversarial Synthesis from Language to Action," in *Proc. of the IEEE International Conference on Robotics and Automation (ICRA)*, May. 2018.

Yoonseon Oh, Kyunghoon Cho, **Yunho Choi**, and Songhwai Oh, "Robust Multi-Layered Sampling-Based Path Planning for Temporal Logic-Based Missions," in *Proc. of the IEEE Conference on Decision and Control (CDC)*, Dec. 2017.

Yunho Choi, Inhwan Hwang, and Songhwai Oh, "Wearable Gesture Control of Agile Micro Quadrotors," in *Proc. of the IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, Nov. 2017.

Honors

Awards and Scholarships

- Graduate Scholarship, Korea Foundation for Advanced Studies — 2017 - Present
- Certificate of Excellent Teaching Assistant in Introduction to IoT·AI·Big Data Course, Seoul National University — 2018
- Summa Cum Laude, Seoul National University — 2017
- Samsung Scholarship for Junior Frontier Leader — 2012

Experience

Research Experience

AI Technology for Guidance of a Mobile Robot to its Goal with Uncertain Maps in Indoor/Outdoor Environments - Ministry of Science and ICT (MSIT)

- Team Lead & Development of a reinforcement learning algorithm for image-goal navigation of a mobile robot — Mar. 2019 - Present

Smart Campus - Samsung Electronics Co., Ltd

- Development of learning-based gesture recognition from sensor data of wearable devices — May 2017 - Apr. 2019

Robot Learning from Demonstrations with Mixed Qualities - National Research Foundation (NRF)

- Development of an inverse optimal control algorithm for demonstrations with mixed qualities — Mar. 2017 - Feb. 2019

Work Experience

Robotics Engineer, Sequor Robotics, Inc.

- Development of Visual SLAM algorithms — Sep. 2023 - Present

Research Intern at Advanced R&D Group, VD Division, Samsung Electronics Co., Ltd

- Development of a vision-based navigation for an RC car — Jul. 2017 - Aug. 2017

Teaching Experience

- Bootcamp for AI Engineers with SOCAR Real-World Data — 2021, 2022
- TA, Deep Reinforcement Learning, Seoul National University — Spring 2019
- TA, Introduction to Deep Learning, SNU Big Data Academy — Fall 2018
- TA, Introduction to IoT·AI·Big Data, Seoul National University — Fall 2018
- TA, Introduction to Intelligent Systems, Seoul National University — Fall 2017

Programming Skills

Language: C++/C, Python, Matlab
Software: PyTorch, TensorFlow, OpenCV